

Sangarshan Reddy Karra

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SUMMARY

Sr. Software Engineer with 8+ years of experience building data-intensive Python backend systems in high-reliability, safety-critical environments, transitioning into Data Engineering. Deep expertise in large-scale data processing (Pandas, PySpark), SQL optimization, ETL pipeline development, and multi-threaded data workflows. Proven track record designing fault-tolerant systems that monitor 2,500+ locomotives in real time, with hands-on experience in Kafka, boto3, cross-system data integration, and pipeline automation across high-volume operational environments. Actively building AI/ML-powered applications using LLMs and cloud-native tools.

PROGRAMMING SKILLS

Programming Languages: Python (primary); Java, JavaScript (familiar)

Data & Pipelines: Pandas, PySpark, Kafka, ETL/ELT Workflows, Oracle PL/SQL, MySQL, PostgreSQL (Query Optimization, Stored Procedures)

Backend & Systems: REST APIs, Flask, Multithreading, XML/XSD Parsing, Object-Oriented Programming, Data Structures

Cloud & Infrastructure: AWS (EC2, S3, boto3, KMS), Docker (local dev, CI/CD, cloud deployment), GitHub Actions / Jenkins, Render.com, RHEL / Linux

AI & LLM Tools: Groq API, Llama 3.3 70B, Prompt Engineering, LLM Application Development

Monitoring & Observability: Zenoss, AWS CloudWatch

Testing & Quality: Unit Testing, TDD, Python unittest, Code Coverage

Dev Tools: Git, GitHub Pages, PyCharm

PROFESSIONAL EXPERIENCE

CSX Technology

Sr. Software Engineer

Jacksonville, FL

May 2019 – Present

- Architected Python backend services powering real-time telemetry monitoring across CSX's fleet of 2,500+ active locomotives, forming the core of the company's federal PTC compliance infrastructure and ensuring 24/7 system availability.
- Owned end-to-end delivery of the onboard locomotive health monitoring service — from authoring the technical design document through cross-functional sign-off with enterprise, QA, and ops teams — shipping to production on schedule with zero critical post-release defects.
- Designed the communication architecture between onboard locomotive systems and enterprise backends, defining XSD message contracts and data formats adopted as the standard across 8 teams.
- Proposed and led approval of the Python 2.6 to 3.12 upgrade across a 100K+ line codebase — presented the business case to engineering leadership, delivered the full up-gradation in under 6 months with zero production incidents, retiring an end-of-life runtime and unblocking adoption of modern Python libraries and tooling.
- Cut data retrieval latency by 25% — from up to 3 minutes to under 2 minutes — by rewriting stored procedures and optimizing SQL queries across high-volume reporting pipelines consumed by 8 downstream teams.
- Built Pandas and PySpark ETL pipelines processing hundreds of thousands of locomotive telemetry records daily, feeding 8 downstream reporting and monitoring systems and enabling data-driven operational decisions across the enterprise.
- Engineered a multithreaded data ingestion and monitoring system concurrently tracking onboard health metrics across 2,500+ locomotives, enabling near real-time anomaly detection and faster incident response for on-call engineering teams.
- Developed and maintained XSD schemas and parsing frameworks governing structured message exchange between onboard and enterprise systems, supporting reliable high-frequency telemetry ingestion at fleet scale.
- Established team-wide unit testing standards with Python unittest, raising codebase coverage from below 60% to 80%+, resulting in measurably fewer production defects across subsequent releases.

- Mentored two engineers from internship through full-time conversion — leading structured code reviews, design sessions, and pair debugging, demonstrating sustained investment in team growth.
- Implemented AWS KMS-based encryption for data at rest and in transit across production services, and served as primary responder for security certificate failures in production — diagnosing and resolving KMS key access issues to restore system operation with minimal downtime.
- Developed integration code interfacing directly with GE and Wabtec onboard locomotive hardware, translating vendor-specific protocols and data formats into standardized messages consumed by CSX's enterprise backend systems.

Guardian Life Insurance

Software/Automation Engineer

Bethlehem, PA
Sep 2017 – Apr 2019

- Automated EC2 inventory reconciliation across 1,000+ instances using boto3 to query AWS APIs and cross-reference against Zenoss monitoring systems, surfacing previously undetected coverage gaps and eliminating up to 5 hours/week of manual audit work.
- Built a Python automation suite replacing recurring manual operational tasks across dev, staging, and production environments, expanding infrastructure monitoring coverage and freeing ops teams for higher-value work.
- Unified Zenoss, CMDB, and AWS data into a single reconciliation pipeline covering 1,000+ EC2 instances, enabling proactive identification of configuration drift and monitoring gaps before they escalated into production incidents.
- Designed and automated infrastructure gap reporting workflows that delivered actionable dashboards to operations teams daily, reducing mean time to detection for unmonitored or misconfigured instances across all environments.
- Automated recurring operational tasks via cron scheduling, improving monitoring coverage consistency across environments and ensuring gap reports ran reliably without manual intervention.

PROJECTS

• AI-Powered Portfolio Chatbot

Apr 2025

Python · Flask · Groq API · Llama 3.3 70B · GitHub Pages · Render.com

- Built a full-stack AI chatbot embedded in a personal portfolio where recruiters can ask natural language questions about experience and skills and receive instant AI-generated responses powered by Llama 3.3 70B via Groq API.
- Engineered a Python Flask REST API with prompt engineering — injecting full resume as system context to scope LLM responses accurately — deployed on Render.com with secure environment variable management and auto-deploy on git push.
- Built a sci-fi themed frontend in vanilla HTML, CSS, and JavaScript featuring a boot sequence, glitch animations, real-time chat UI, and scroll-triggered effects — hosted on GitHub Pages with zero infrastructure cost.
- Diagnosed and resolved 6 production issues end-to-end: API key security incident with git history rewrite, Python environment conflicts, port collisions, LLM model deprecation, and cloud deployment failures.

• Interactive Sci-Fi Portfolio Website

Apr 2025

HTML · CSS · JavaScript · GitHub Pages

- Designed and built a classified-dossier themed interactive resume with animated skill bars, typewriter effects, HUD aesthetics, canvas-based grid background, and scroll-triggered reveals — deployed on GitHub Pages.

EDUCATION

University of Missouri Kansas City

Master of Science in Electrical Engineering

Kakatiya Institute of Technology & Science

Bachelor of Technology in Electronics and Communication Engineering

Kansas City, MO
January 2016 – July 2017

Warangal, India
August 2011 – May 2015